Homework 4

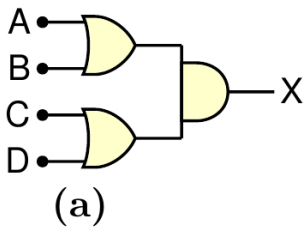
COSC330/PHYS306 · Chapters 5 and 6

Prof. Jordan C. Hanson

Problem 1 [Section 5-1, Exercise 6]

Question: *Develop the truth table for each circuit in Figure 5–56.*

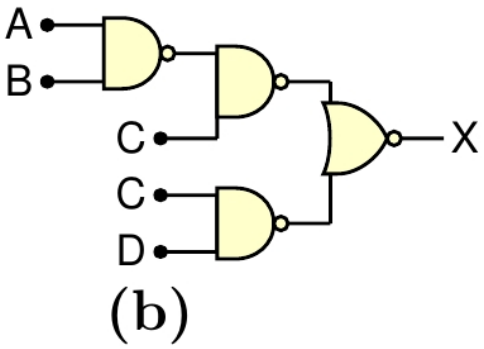
Circuit (a)



Answer: X = (A + B)(C + D)

A	B	C	D	A+B	C+D	X
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

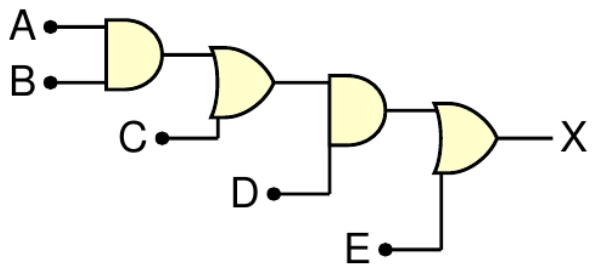
Circuit (b)



Answer: X = (AB)'·C·D

A	B	C	D	(AB)'	((AB)'C)'	(CD)'	X
0	0	0	0	1	1	1	0
0	0	0	1	1	1	1	0
0	0	1	0	1	0	1	0
0	0	1	1	1	0	0	1
0	1	0	0	1	1	1	0
0	1	0	1	1	1	1	0
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	1	1	0
1	0	0	1	1	1	1	0
1	0	1	0	1	0	1	0
1	0	1	1	1	0	0	1
1	1	0	0	0	1	1	0
1	1	0	1	0	1	1	0
1	1	1	0	0	1	1	0
1	1	1	1	0	1	0	0

Circuit (c)

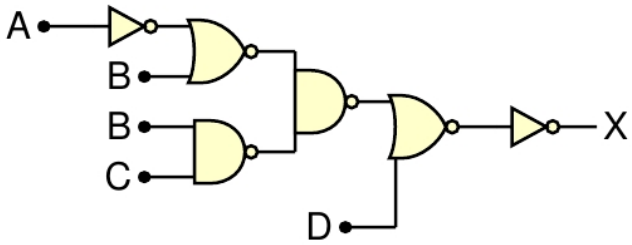


(c)

Answer: X = (AB + C)D + E = ABD + CD + E

A	B	C	D	E	AB	AB+C	(AB+C)D	X
0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	0	0	0
0	0	0	1	1	0	0	0	1
0	0	1	0	0	0	1	0	0
0	0	1	0	1	0	1	0	1
0	0	1	1	0	0	1	1	1
0	0	1	1	1	0	1	1	1
0	1	0	0	0	0	0	0	0
0	1	0	0	1	0	0	0	1
0	1	0	1	0	0	0	0	0
0	1	0	1	1	0	0	0	1
0	1	1	0	0	0	1	0	0
0	1	1	0	1	0	1	0	1
0	1	1	1	0	0	1	1	1
0	1	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	0
1	0	0	1	1	0	0	0	1
1	0	1	0	0	0	1	0	0
1	0	1	0	1	0	1	0	1
1	0	1	1	0	0	1	1	1
1	0	1	1	1	0	1	1	1
1	1	0	0	0	1	1	0	0
1	1	0	0	1	1	1	0	1
1	1	0	1	0	1	1	1	1
1	1	0	1	1	1	1	1	1
1	1	1	0	0	1	1	0	0
1	1	1	0	1	1	1	0	1
1	1	1	1	0	1	1	1	1
1	1	1	1	1	1	1	1	1

Circuit (d)

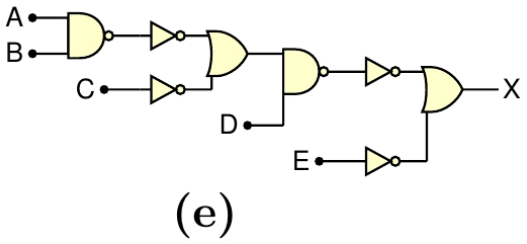


(d)

Answer: X = A' + B + D

A	B	C	D	(A'+B)'	(BC)'	G3 (NAND)	G4 (NOR)	X
0	0	0	0	0	1	1	0	1
0	0	0	1	0	1	1	0	1
0	0	1	0	0	1	1	0	1
0	0	1	1	0	1	1	0	1
0	1	0	0	0	1	1	0	1
0	1	0	1	0	1	1	0	1
0	1	1	0	0	0	1	0	1
0	1	1	1	0	0	1	0	1
1	0	0	0	1	1	0	1	0
1	0	0	1	1	1	0	0	1
1	0	1	0	1	1	0	1	0
1	0	1	1	1	1	0	0	1
1	1	0	0	0	1	1	0	1
1	1	0	1	0	1	1	0	1
1	1	1	0	0	0	1	0	1
1	1	1	1	0	0	1	0	1

Circuit (e)

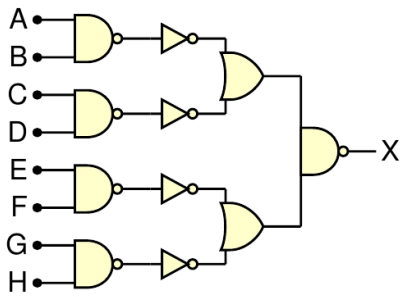


(e)

Answer: X = (AB + C')D + E'

A	B	C	D	E	AB	AB+C'	(AB+C')D	X
0	0	0	0	0	0	1	0	1
0	0	0	0	1	0	1	0	0
0	0	0	1	0	0	1	1	1
0	0	0	1	1	0	1	1	1
0	0	1	0	0	0	0	0	1
0	0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0	1
0	0	1	1	1	0	0	0	0
0	1	0	0	0	0	1	0	1
0	1	0	0	1	0	1	0	0
0	1	0	1	0	0	1	1	1
0	1	0	1	1	0	1	1	1
0	1	1	0	0	0	0	0	1
0	1	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0
1	0	0	0	0	0	1	0	1
1	0	0	0	1	0	1	0	0
1	0	0	1	0	0	1	1	1
1	0	0	1	1	0	1	1	1
1	0	1	0	0	0	0	0	1
1	0	1	0	1	0	0	0	0
1	0	1	1	0	0	0	0	1
1	0	1	1	1	0	0	0	0
1	1	0	0	0	1	1	0	1
1	1	0	0	1	1	1	0	0
1	1	0	1	0	1	1	1	1
1	1	0	1	1	1	1	1	1
1	1	1	0	0	1	1	0	1
1	1	1	0	1	1	1	0	0
1	1	1	1	0	1	1	1	1
1	1	1	1	1	1	1	1	1

Circuit (f)



(f)

FIGURE 5–56

Answer: X = [(AB + CD)(EF + GH)]'

ABCD \ EFGH	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0000	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0001	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0010	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0011	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
0100	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0101	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0110	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0111	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1000	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1001	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1010	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1011	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1100	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1101	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1110	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1111	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0

Problem 2 [Section 5-2, Exercise 13]

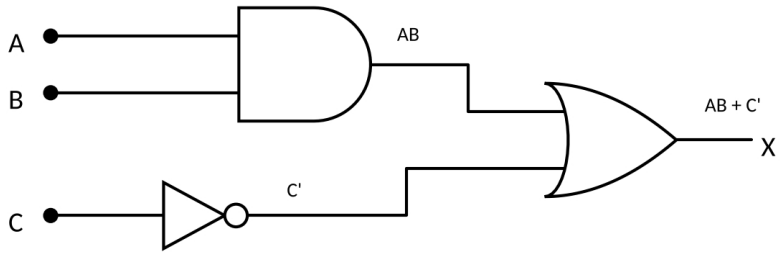
Question: *Implement a logic circuit for the truth table in Table 5–8.*

Inputs			Output
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

TABLE 5–8

Answer: X = AB + C'

A	B	C	AB	C'	X = AB + C'
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	1



Problem 3 [Section 5-3, Exercise 20]

Question: *Implement the logic circuits in Figure 5–54 using only NAND gates.*

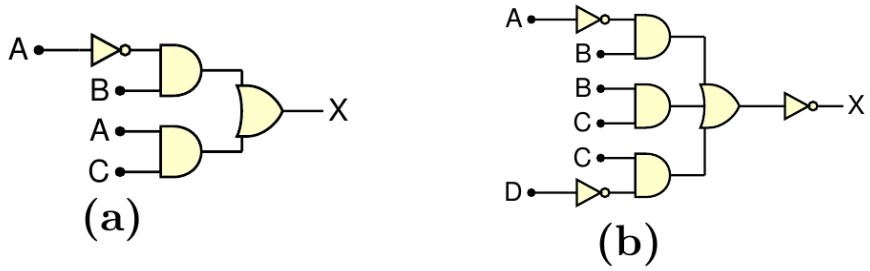
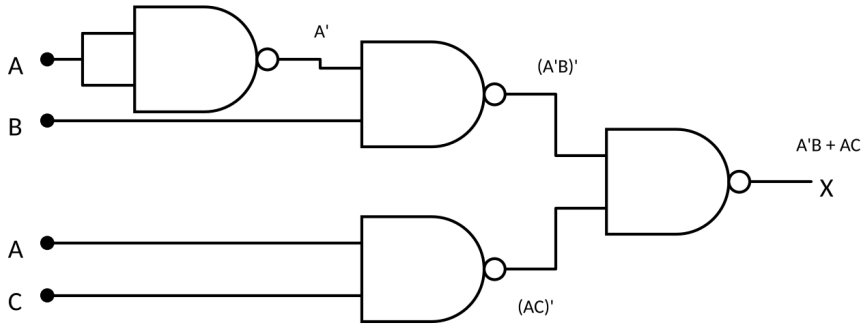
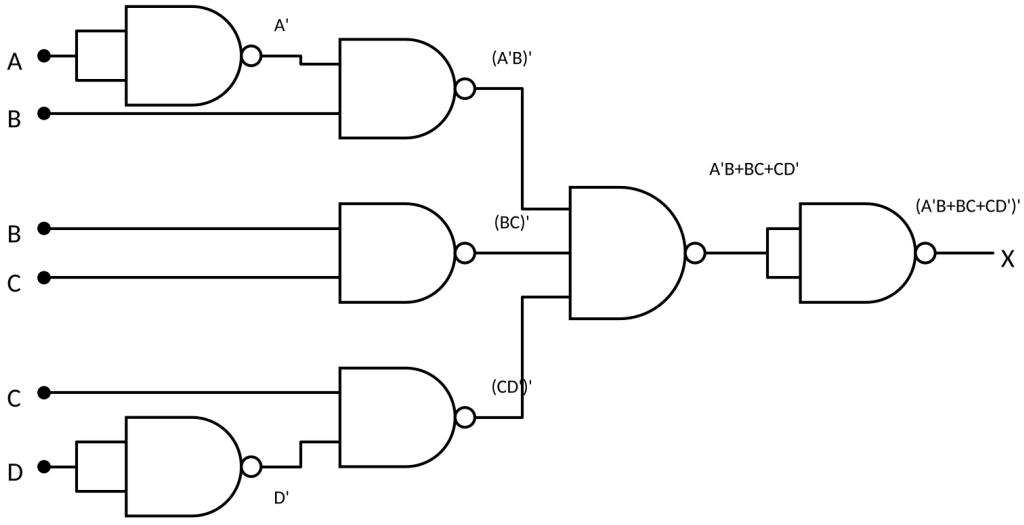


FIGURE 5–54

Answer: X = [(A'B)'·(AC)']' = A'B + AC (4 NAND gates)



Answer: X = (A'B + BC + CD')' (7 NAND gates)



Problem 4 [Section 6-2, Exercise 4]

Question: *For the parallel adder in Figure 6–69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers.*

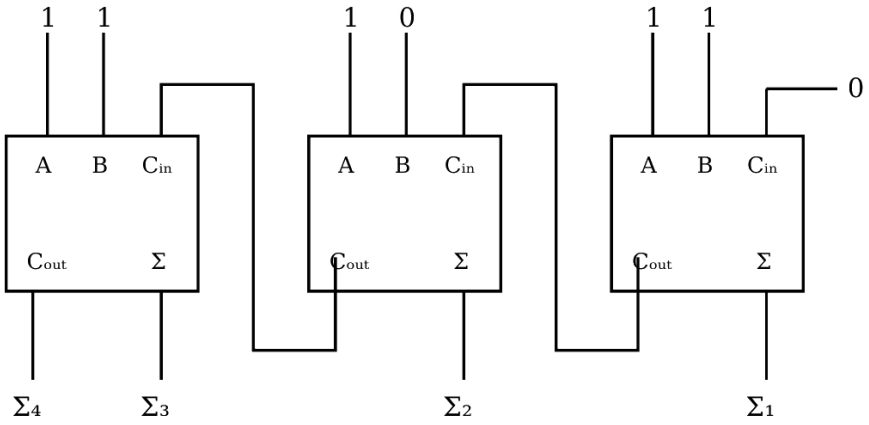


FIGURE 6–69

Stage	A	B	Cin	Σ	Cout
1 (right)	1	1	0	Σ1 = 0	1
2 (middle)	1	0	1	Σ2 = 0	1
3 (left)	1	1	1	Σ3 = 1	1 = Σ4

Sum = 1100

Longhand verification (A = 111 = 7, B = 101 = 5):

```
carries:  1 1
        1 1 1    (7)
      + 1 0 1    (5)
      -------
      1 1 0 0    (12)
```

Problem 5 [Section 6-2, Exercise 8]

Question: *The input waveforms in Figure 6–72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.*

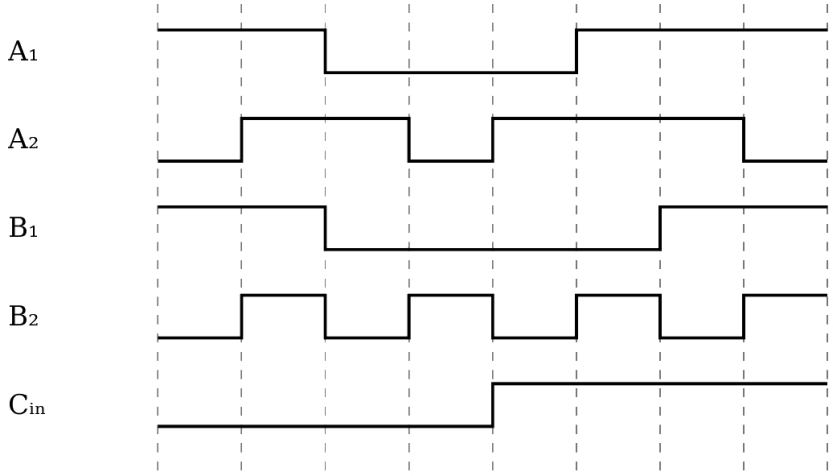
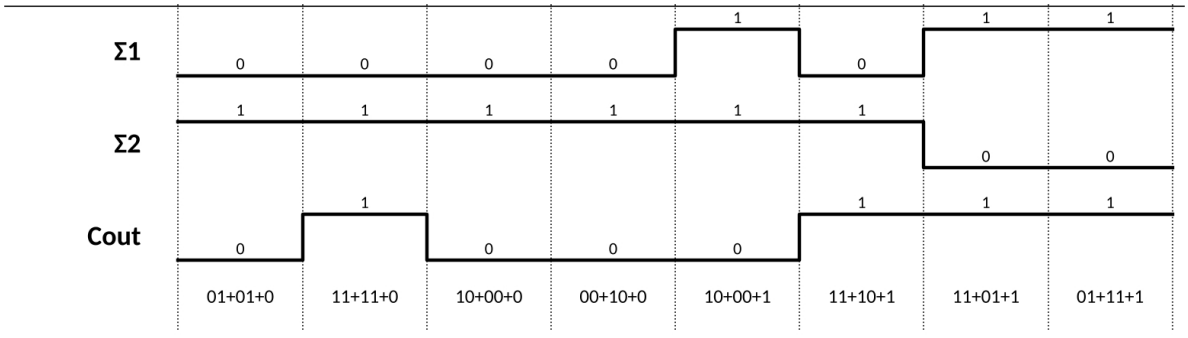


FIGURE 6–72

Answer: Σ1 = 0 0 0 0 1 0 1 1, Σ2 = 1 1 1 1 1 1 0 0, Cout = 0 1 0 0 0 1 1 1

Interval	A2A1	B2B1	Cin	Sum	Cout	Σ2	Σ1
t1	01	01	0	2	0	1	0
t2	11	11	0	6	1	1	0
t3	10	00	0	2	0	1	0
t4	00	10	0	2	0	1	0
t5	10	00	1	3	0	1	1
t6	11	10	1	6	1	1	0
t7	11	01	1	5	1	0	1
t8	01	11	1	5	1	0	1



Problem 6 [Section 6-4, Exercise 14]

Question: *For the 4-bit comparator in Figure 6–74, plot each output waveform for the inputs shown. The outputs are active-HIGH.*

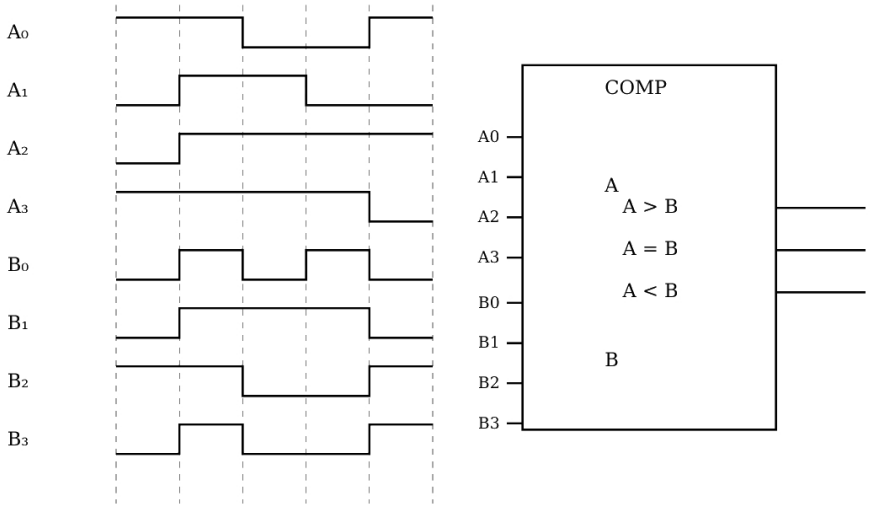
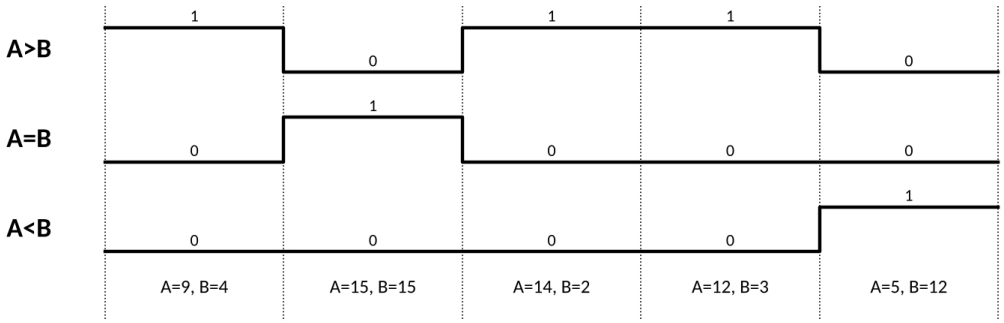


FIGURE 6–74

Answer: A > B HIGH in t1, t3, t4; A = B HIGH in t2; A < B HIGH in t5

Interval	A3A2A1A0	A	B3B2B1B0	B	A > B	A = B	A < B
t1	1001	9	0100	4	1	0	0
t2	1111	15	1111	15	0	1	0
t3	1110	14	0010	2	1	0	0
t4	1100	12	0011	3	1	0	0
t5	0101	5	1100	12	0	0	1



Problem 7 [Section 6-5, Exercise 16]

Question: *When a HIGH is on the output of each of the decoding gates in Figure 6–75, what is the binary code appearing on the inputs? The MSB is A3.*

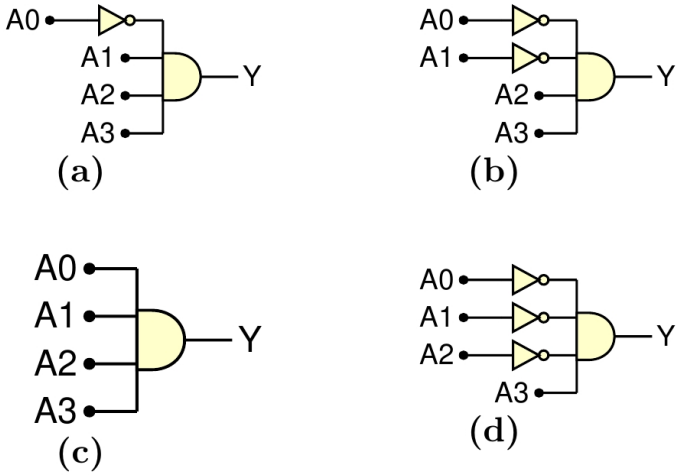


FIGURE 6–75

Answer: (a) 1110 (b) 1100 (c) 1111 (d) 1000

Problem 8 [Section 6-8, Exercise 28]

Question: *For the multiplexer in Figure 6–79, determine the output for the following input states: D0 = 0, D1 = 1, D2 = 1, D3 = 0, S0 = 1, S1 = 0.*

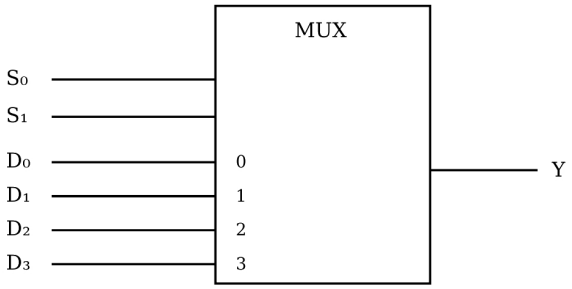


FIGURE 6–79

Answer: Y = 1